Student Performance Analytics System

Insights Report

Prepared By

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1. Project Overview

The Student Performance Analytics System was developed to analyze student academic performance and identify factors that influence learning outcomes. The project involved data cleaning, exploratory data analysis (EDA), feature engineering, and dashboard development using Python and Google Colab.

The objective was to transform raw student data into meaningful insights that can support educational decision-making.

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2. Dataset Summary

The dataset contains information related to student performance, including:

- Study Hours
- Internal Marks
- Final Marks
- Assignment Scores
- Attendance
- Family Income
- Internet Access
- Extra Activities
- Placement Status

A total of 10,000 student records were analyzed.

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3. Data Preparation

The following preprocessing steps were performed:

- Removed duplicate records
- Checked for missing values
- Standardized data formats
- Created additional analytical features
- Prepared data for visualization and reporting

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4. Key Findings

Study Hours and Academic Performance

A strong positive relationship was observed between study hours and final marks. Students who dedicated more time to studying generally achieved higher academic performance.

Attendance Impact

Attendance showed a positive influence on student results. Students with higher attendance percentages consistently performed better than those with lower attendance.

Internet Access

Students with internet access demonstrated slightly better academic performance, indicating the importance of digital learning resources.

Family Income

Family income showed a moderate impact on performance. While higher-income students tended to perform better on average, study habits remained the most influential factor.

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5. Risk Analysis

Students were classified into risk categories based on their final marks:

- At Risk: Marks below 40
- Safe: Marks 40 and above

This categorization helps identify students who may require additional academic support and intervention.

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6. Performance Classification

Students were grouped into the following categories:

- Excellent (80 and above)
- Good (60–79)
- Average (40–59)
- Poor (Below 40)

This classification provides a clear understanding of overall student distribution and performance levels.

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7. Dashboard Development

An interactive dashboard was developed using Streamlit to provide:

- Student performance overview
- Performance level filtering
- Attendance and marks analysis
- Correlation heatmap
- Downloadable filtered data

The dashboard enables users to explore data and insights efficiently.

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8. Conclusion

The analysis indicates that study hours and attendance are the strongest factors influencing student success. Internet access and socioeconomic conditions also contribute to academic outcomes but have a smaller impact compared to consistent study habits.

This project demonstrates the effective use of data analytics for identifying performance trends, supporting student success, and enabling data-driven educational decisions.

Tools & Technologies

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Streamlit
- Google Colab

End of Report